When Should an LLM Issue Triager Stay Silent? Calibration, Drift, Rewording and Evidence in Issue Triage

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Abstract

Issue-labeling services now apply LLM labels to open-source issues automatically. We evaluate one as a deployed, selective service: Qwen2.5-7B, zero-shot and with retrieved examples, against classical classifiers, on the NLBSE'24 benchmark and on 1,430 issues the same five repositories received in 2025–26, after the model's training cutoff. The classical TF-IDF classifier is the most accurate on the benchmark (76.2%) but falls to 67.5% on 2025–26 issues and 60.3% on a new repository; the LLM is stable (70.9%, 75.3%, 68.0%). Raw LLM probabilities are badly overconfident. A confidence threshold tuned for 10% error among auto-labeled issues overshoots in every condition (14.9% in distribution, 20.5% on a new repository); a risk-controlled threshold keeps the target in distribution and a year later but auto-labels only 7% of issues, and no threshold carries over to a new repository. A one-sentence newcomer apology flips 16.2% of the few-shot LLM's labels, mostly toward question, and non-native English costs the zero-shot LLM up to 10 points; calibrated deferral keeps almost all such flips from being auto-applied. The LLM's evidence quotes are genuine (93% verbatim) but deleting them moves its decision little more than deleting random text. Input novelty detects a new repository within 11 requests, while the model's own confidence rarely notices. A record/replay regression suite catches injected bugs in calibration, retrieval and preprocessing.

Keywords

issue triage, large language models, selective prediction, calibration, robustness, explanations, ML in production

1 Introduction

Open-source projects receive more issues than maintainers can read, and automatic labeling is one of the most requested forms of help. GitHub now ships an LLM-based labeler as an Action [8], and research has moved from fine-tuned classifiers to LLMs [3, 5]. Benchmarks report how often a labeler is right on a fixed snapshot [14]. A deployed labeler faces different questions. Which labels should it apply on its own, and which should it leave for a person? Does its promised error rate hold a year later, or on a repository it has never seen? Does it treat a report differently because the reporter writes in non-native English or apologizes for being new? And when it explains a decision, is the explanation what actually drove it?

We study these questions with a triage service built around Qwen2.5-7B, running locally, on the NLBSE'24 benchmark (3,000 issues from five large repositories, 2016–2023) and on 1,430 issues the same repositories received in 2025–2026, after the model's training cutoff. We compare the LLM, zero-shot and with retrieved examples, against classical classifiers trained on the repositories' history, and ask:

- **RQ1** How do classical and LLM triagers compare in distribution, a year later, and on a new repository?
- **RQ2** Can calibrated deferral keep the error among auto-applied labels at a target (10%), and does that promise survive drift?
- **RQ3** Do meaning-preserving changes in how an issue is written (no stack trace, no template, a newcomer apology, non-native English) change the label?
- **RQ4** Are the evidence quotes the LLM gives faithful to its decision?
- **RQ5** Which logged signal notices that the stream of issues has changed?

The answers are mixed in useful ways. A TF-IDF classifier trained on the history is the most accurate in distribution (76.2%) but drops to 67.5% on 2025–26 issues and 60.3% on a new repository; the LLM is less accurate in distribution (70.9%) but stable (75.3% and 68.0%). Raw LLM label probabilities are badly overconfident (fitted temperatures of 8–10). A confidence threshold tuned to 10% error on calibration data overshoots in every condition (14.9% in distribution, 20.5% on a new repository); a risk-controlled threshold keeps the promise in distribution and a year later but auto-labels only 7% of issues. A one-sentence newcomer apology flips 16.2% of the few-shot LLM's labels, mostly toward *question*, yet calibrated deferral keeps almost all flips from being auto-applied. The LLM's evidence quotes are real quotes, but deleting them moves its decision little more than deleting random text. A simple novelty signal, distance to the closest labeled history issue, detects a new repository within 11 requests; the model's own confidence rarely notices.

Contributions. (1) An evaluation of LLM issue triage as a deployed, selective service, including a post-cutoff 2025–26 set from the benchmark's own repositories; (2) evidence that common calibration recipes over-promise and that rigorous risk control is affordable only with far more calibration data; (3) behavioral tests showing a newcomer-apology sensitivity that in-context examples amplify; (4) a faithfulness test of LLM triage explanations; and (5) monitoring and record/replay regression tests for the service, all on one laptop.

2 Related Work

Issue classification. Ticket Tagger predicts GitHub issue types from text with fastText [11]. The NLBSE tool competitions turned the task into a shared benchmark; the 2024 edition provides 3,000 balanced issues from five repositories and a SetFit baseline [14]. Colavito et al. benchmark 22 decoder LLMs against fine-tuned encoders and find zero-shot LLMs less consistent across datasets [5];

Aracena et al. compare GPT-4o and DeepSeek-R1 on the same data [3]. GitHub now ships an LLM-based issue labeler as an Action [8]. These works report accuracy on a fixed benchmark; we ask how a deployed labeler should decide when *not* to label, and what happens after the benchmark period.

Calibration and selective prediction. Modern networks are overconfident, and temperature scaling is a simple fix [9]. Selective classification abstains below a confidence threshold chosen for a target risk [7]; split conformal prediction gives distribution-free coverage under exchangeability [1]. We test whether these guarantees survive the shifts a triage service meets in production.

Behavioral testing, fairness and explanations. CheckList tests NLP models with meaning-preserving perturbations [17]. LLMs answer users with lower English proficiency less accurately [15], which motivates testing whether a triager treats non-native reporters differently. ERASER measures whether rationales are faithful by deleting them [6]; we apply the same idea to the evidence quotes an LLM triager gives.

ML in production. Monitoring, calibration under drift, and regression tests are part of production readiness [4, 12].

3 Study Design

3.1 The triage service

The service receives a new GitHub issue and either applies one label (*bug*, *feature*, *question*) or leaves the issue for a maintainer. Bodies are preprocessed as a deployed service would: HTML comments left by issue templates are removed and the body is cut to 1,500 characters. We compare four classifiers. Two are classical: TF-IDF features with logistic regression, and `bge-small-en-v1.5` [18] embeddings with logistic regression. Two use Qwen2.5-7B-Instruct [16] (4-bit, MLX [10], one laptop): *zero-shot* with label definitions, and *few-shot*, which adds the six most similar labeled issues from the same repository, retrieved by embedding similarity. For the LLMs we read the next-token distribution after the prompt and take the probability mass on each label word, renormalized over the three labels; this costs one prefill pass and needs no verbalized confidence.

3.2 Data and conditions

NLBSE'24 [14] supplies 3,000 issues (react, tensorflow, vscode, bitcoin, opencv; 2016–2023; 100 per repository and label in each half). We split its training half into a labeled history of 1,005 issues (training data for the classical models and retrieval pool for the few-shot LLM) and a calibration split of 495, and use its full test half (1,500 issues). We evaluate three conditions:

In-distribution: the NLBSE'24 test set.

Fresh 2025–26: issues created between January 2025 and October 2026 in the same five repositories, collected with the GitHub API and mapped to the three labels through each repository's own label names (1,430 single-type issues; a balanced sample of up to 40 per repository and label, 514 issues). History and calibration stay in 2016–2023, as for a service deployed and not retrained; the period also lies after the model's training cutoff, which removes benchmark leakage.

New repository: 750 test issues triaged as if their repository had just been onboarded: the few-shot model retrieves examples only from the other four repositories and the classical models are trained without the target repository.

3.3 Selective automation

Each classifier's scores are temperature-scaled [9] on the calibration split. We compare five policies: *auto-all*; a *naive threshold* on calibrated confidence, the lowest one whose auto-labeled calibration issues have at most 10% errors [7]; a *risk-controlled threshold* chosen by Learn-then-Test [2], which tests thresholds from the most confident down and accepts each while a 95% Clopper–Pearson upper bound on its calibration error stays at or below 10%; *conformal*, which auto-labels only when the split-conformal prediction set at α=0.1 has one label [1]; and *agreement*, which auto-labels only when the few-shot LLM and the embedding classifier agree. Thresholds and conformal quantiles are set once on the calibration split and applied unchanged in every condition.

3.4 Rewording tests

From the test set we draw 210 probe issues (14 per repository and label) and create five meaning-preserving variants each: *no code* (fenced blocks and stack-trace lines removed), *no template* (template headings and checklists removed), *newcomer* (a sentence saying the reporter is new and apologizing for a possibly silly question), *non-native* (the prose rewritten as by a non-native English speaker; code blocks are masked during rewriting and restored), and *non-native + no code*. The non-native rewrite is produced by Llama-3.2-3B [13], a different model family from the classifier under test, and kept only if every code block is restored, the length stays within 0.5–1.8×, and at least 80% of technical tokens survive. A label flip between an issue and its variant is a robustness failure, because the issue type is unchanged by construction.

3.5 Explanations

For the probe issues the few-shot model is also asked for its label and two exact quotes from the issue as evidence. We check whether the quotes occur verbatim and run a deletion test [6]: the issue is re-scored, with the same retrieved examples, after deleting the quoted text and, as a control, after deleting random spans of the same total length (two seeds).

3.6 Statistics

Intervals are percentile bootstraps over issues (2,000 resamples). Monitoring follows the protocol of Section 4.5.

4 Results

4.1 RQ1: Accuracy in and out of distribution

Table 1. In distribution, TF-IDF with logistic regression, trained on only 1,005 labeled issues, is the most accurate approach (76.2%, 95% CI [74.0, 78.3]); the few-shot LLM reaches 70.9% [68.6, 73.3] and zero-shot 68.5%. Out of distribution the order reverses. On 2025–26 issues TF-IDF drops to 67.5% and its cross-repository F1 from 0.760 to 0.584, while the few-shot LLM rises to 75.3%. On a new repository TF-IDF and the embedding classifier fall to 60.3% and 58.8%; the

Table 1: Accuracy (%) by condition and cross-repository F1. Tokens: prompt tokens per issue.

Approach	NLBSE'24	2025–26	New repo	Tokens
TF-IDF + LR	76.2	67.5	60.3	–
Embedding + LR	69.6	68.1	58.8	–
LLM zero-shot	68.5	71.2	68.4	0.4k
LLM few-shot	70.9	75.3	68.0	1.1k
Cross-repository F1 (NLBSE ranking metric)				
TF-IDF + LR	0.760	0.584	0.589	
Embedding + LR	0.694	0.640	0.576	
LLM zero-shot	0.663	0.650	0.661	
LLM few-shot	0.694	0.682	0.657	

LLMs lose at most 3 points (68.0%, 68.4%). The classical models learn repository- and period-specific vocabulary; the LLM relies on general knowledge of what bugs, features and questions look like.

The 2025–26 issues also serve as a leakage control: they postdate the model's training data, and the LLM is *more* accurate on them than on the 2016–2023 benchmark, so we see no sign that benchmark contamination inflates LLM results here. The fresh data show a different kind of drift instead: label usage. React, for example, labeled 200 bugs but only 9 feature requests and 5 questions in 2025–26, since it now routes questions elsewhere, so a labeler tuned to the benchmark's balanced classes faces a very different prior.

4.2 RQ2: Does calibrated deferral keep its promise?

Raw LLM label probabilities are almost always 0 or 1; temperature scaling on the calibration split needs T=7.6 (few-shot) and T=10.1 (zero-shot) and lowers expected calibration error from 0.23–0.30 to 0.05–0.08. The classical models are underconfident ($T \approx 0.6$).

Table 2 and Figure 1 show what happens when the scores are used to decide which issues to label automatically. The *naive threshold* — the lowest threshold at which the auto-labeled calibration issues contain at most 10% errors — overshoots the target in every condition, even in distribution (TF-IDF 12.5%, few-shot LLM 14.9%), and badly under shift (TF-IDF 26.3%, LLM 20.5% on a new repository). Choosing a threshold on 495 calibration issues is optimistic by construction: the lowest feasible threshold sits where calibration noise happens to look good. The *risk-controlled* threshold keeps the promise in distribution (3.9–5.9% risk) and on 2025–26 issues (0–9.4%), but it pays with coverage: the few-shot LLM auto-labels only 6.9% of issues, the zero-shot LLM none, and TF-IDF 15.6%. On a new repository it fails for every classifier that auto-labels anything (11.1–18.2%), because the guarantee assumes calibration and deployment data are exchangeable. In hindsight, the test data would have allowed auto-labeling 19.1% of issues at 10% error with the few-shot LLM and 45.5% with TF-IDF; on a new repository no threshold reaches 10% for any trained classifier. With 495 calibration issues, the naive rule overshoots that point and the rigorous rule stops far short of it. Conformal prediction sets cover the true label in 87.5–89.3% of in-distribution cases (target 90%); on a new repository the classical classifiers fall to 80.0–83.5%, while the LLMs stay at 86–89% in every condition. An agreement gate between the LLM and the embedding classifier labels about two thirds of issues at 17.7–26.4% risk, far from the target.

4.3 RQ3: Does the way an issue is written change its label?

Table 3 and Figure 2. TF-IDF is the most fragile: removing code, removing template scaffolding or rewriting in non-native English each flips 13–17% of its labels. The LLMs are more robust to style and structure, with one striking exception: a single sentence in which the reporter says they are new and apologizes for a possibly silly question flips 16.2% of the few-shot LLM's labels, 14.8% toward *question*, against 5.7% for zero-shot. The effect does not come through retrieval: 85% of the retrieved examples are unchanged and the share of question examples stays at 0.33. In-context examples make the model more sensitive to the cue itself. Non-native English costs the zero-shot LLM 6.9 points of accuracy, and 10.1 points when the stack trace is also missing, which matters for reporters who rely on logs because their English is limited.

Deferral absorbs almost all of this: with the naive threshold, the share of variants where both the original and the rewritten issue would be auto-labeled with different labels is 0–1.3% for every classifier. Flips happen among issues the service was already unsure about.

4.4 RQ4: Are the explanations faithful?

Asked for a label and two exact quotes as evidence, the few-shot LLM quoted the issue in 178 of 210 cases; 93.1% of its 331 quotes occur verbatim, and the explained label matches the scored label in 92.9% of issues. Faithfulness is weaker. On 164 issues with locatable quotes (on average 130 of 921 body characters), deleting the quoted evidence lowered the calibrated probability of the predicted label by 0.025, against 0.014 for deleting random text of the same length (Wilcoxon p=0.115); it changed the label in 6.1% of issues against 3.0% for random deletion, and lowered confidence more than random deletion in only 55.5% of issues. The quotes are plausible and real, but the decision is spread across the issue and the retrieved examples; they should not be presented as *the* reason.

4.5 RQ5: Which signal notices a changed stream?

We simulate 300 streams per shift: 400 in-distribution requests, then 200 from 2025–26, from a repository without history, or in non-native English. A signal alarms when its rate over the last 50 requests exceeds the in-distribution mean plus three standard deviations. Table 4 and Figure 3: *novelty*, one minus the cosine similarity to the closest labeled history issue the service may retrieve from, detects a new repository a median of 11 requests after the shift and 2025–26 traffic after 38 (98% of runs), with no false alarms. The model's own signals — share handed to a human, low confidence, disagreement with the embedding classifier — detect these shifts in under a third of runs, and the predicted-label distribution in at most half. No signal reliably notices the non-native stream. Model-confidence monitoring alone would miss drift that input monitoring catches immediately.

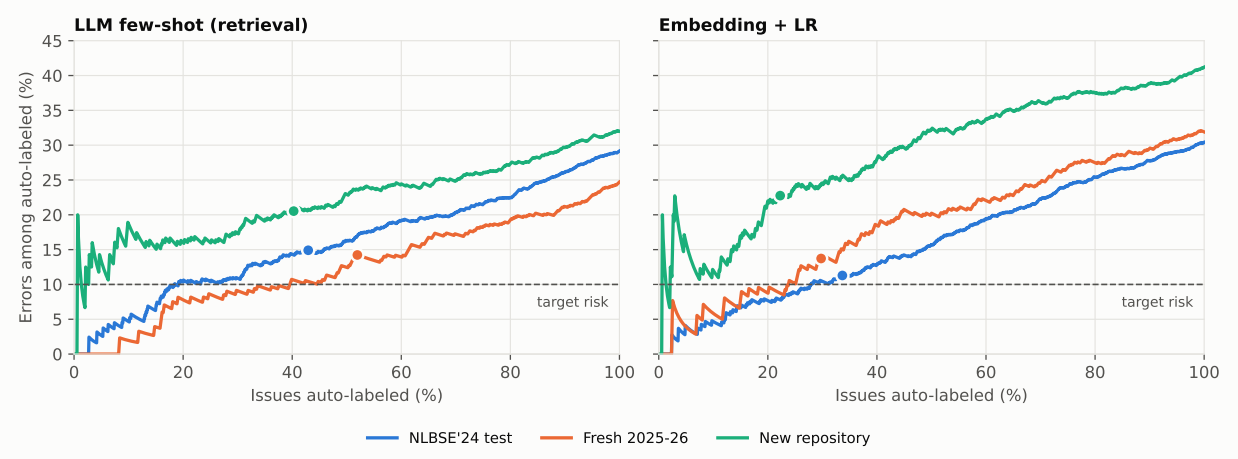


Figure 1: Risk–coverage curves: error among the most confident issues as more are auto-labeled. Dots: the naive threshold calibrated for 10% error. The dashed line is the target.

Table 2: Selective automation: coverage / risk (%), i.e., share auto-labeled / errors among them. Target risk: 10%. All thresholds are set on the calibration split.

Approach	Policy	NLBSE'24	2025–26	New repo
TF-IDF + LR	auto-all	100.0 / 23.8	100.0 / 32.5	100.0 / 39.7
	naive threshold	54.9 / 12.5	48.8 / 19.1	33.5 / 26.3
	risk-controlled	15.6 / 4.3	12.5 / 9.4	1.5 / 18.2
	conformal	61.9 / 14.5	60.7 / 23.7	46.5 / 28.9
Embedding + LR	auto-all	100.0 / 30.4	100.0 / 31.9	100.0 / 41.2
	naive threshold	33.7 / 11.3	29.8 / 13.7	22.3 / 22.8
	risk-controlled	13.5 / 5.9	11.5 / 5.1	8.0 / 11.7
	conformal	47.7 / 15.7	44.7 / 18.3	38.5 / 28.7
LLM zero-shot	auto-all	100.0 / 31.5	100.0 / 28.8	100.0 / 31.6
	naive threshold	13.1 / 12.2	24.9 / 7.8	10.9 / 14.6
	risk-controlled	0.0 / –	0.0 / –	0.0 / –
	conformal	29.5 / 17.8	39.3 / 11.4	27.1 / 17.2
LLM few-shot	auto-all	100.0 / 29.1	100.0 / 24.7	100.0 / 32.0
	naive threshold	42.9 / 14.9	51.9 / 14.2	40.3 / 20.5
	risk-controlled	6.9 / 3.9	8.0 / 0.0	1.2 / 11.1
	conformal	52.8 / 17.9	59.5 / 16.3	50.5 / 22.7
LLM few-shot + Emb.	agreement	67.0 / 17.8	71.4 / 17.7	63.6 / 26.4

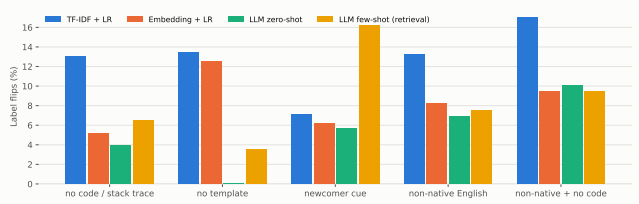


Figure 2: Label flips under rewording.

4.6 Regression tests

A 40-case suite (issues auto-labeled correctly, wrong predictions that must be handed off, and rewrites that must keep their label) replays the full pipeline from recorded model scores in about 6 seconds. Injecting plausible bugs and replaying live: removing temperature scaling fails 14 cases (every must-hand-off case is then auto-labeled), retrieving examples from other repositories fails 10, cutting the issue body to 300 characters fails 6, dropping retrieval fails 5, and removing the label definitions from the prompt fails only 1, a gap in the suite.

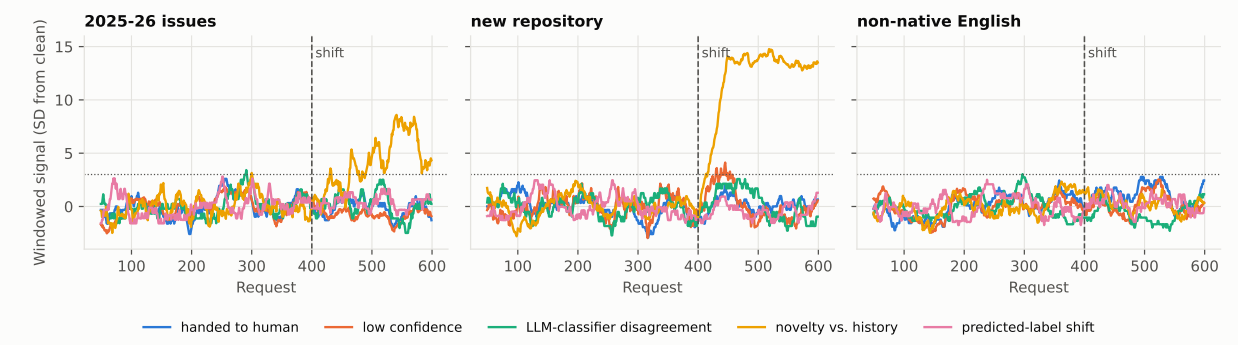


Figure 3: Windowed signals in one simulated stream per shift, standardized against the clean segment; the dotted line is the alarm level.

Table 3: Label flips (%) under meaning-preserving rewording of 210 probe issues (*n*: applicable variants), and accuracy change.

Approach	No code	No templ.	Newcomer	Non-native	NN + no code
n	77	112	210	159	159
TF-IDF + LR	13.0	13.4	7.1	13.2	17.0
Embedding + LR	5.2	12.5	6.2	8.2	9.4
LLM zero-shot	3.9	0.0	5.7	6.9	10.1
LLM few-shot	6.5	3.6	16.2	7.5	9.4
Accuracy change, original → variant (points)					
TF-IDF + LR	-1.3	+4.5	-2.9	-0.6	-3.8
Embedding + LR	-5.2	+0.0	+2.9	-2.5	-3.8
LLM zero-shot	-3.9	+0.0	-2.4	-6.9	-10.1
LLM few-shot	+0.0	-1.8	-3.3	-4.4	-5.0

Table 4: Detection delay after a shift: median requests to the first alarm (share of 300 runs detected); –: detected in fewer than half.

Shift	Handoff	Low conf.	Disagree	Novelty	Label shift
2025–26 issues	–	–	–	38 (98%)	–
New repository	–	–	–	11 (100%)	–
Non-native	–	–	–	–	–

5 Lessons for Practitioners

Benchmark accuracy is the wrong selection criterion. The most accurate classifier on the benchmark was the least accurate a year later and on a new repository. Evaluate a labeler on issues from after its training and calibration data, and on a held-out repository, before choosing it.

A calibrated threshold is not an error guarantee. Tuning the threshold to the target on a few hundred calibration issues over-promised in every condition. Use a risk-controlling procedure, and budget for thousands of labeled calibration issues if the target should be tight and coverage useful. Recalibrate when a repository is onboarded; no threshold carried over.

Defer, and the rewording problem mostly goes away. Labels flip under harmless rewording, and a newcomer's apology pushes the LLM toward *question*, but those flips concentrate where the service is unsure. Deferral is a fairness mechanism as much as an accuracy one, provided that deferred issues are actually reviewed.

Show evidence as context, not as cause. The LLM's quotes are genuine but only weakly faithful; a triage UI should present them as relevant passages, not as the reason for the label.

Monitor inputs, not only the model. A nearest-neighbor novelty score over the labeled history caught a new repository within a dozen requests; the model's confidence did not.

6 Threats to Validity

Construct. Labels are the repositories' own type labels, which mix issue content with project policy (React routes questions elsewhere). The 2025–26 set maps each repository's label names to the three types; issues with several type labels are excluded. Rewrites are synthetic: the non-native rewrite was validated automatically (code blocks restored, length and technical tokens preserved; 161 of 210 valid) and spot-checked by hand, and occasionally alters template fields.

Internal. The calibration split comes from the benchmark's training half and the test set from its test half, as in the competition; thresholds are fixed before evaluation. Probabilities are read from the first output token; a different prompt or label wording would change them.

External. One LLM, five large repositories, three labels. Smaller projects, other label sets, and stronger models (including fine-tuned encoders such as the SetFit competition baseline) may behave differently. All runs used one laptop.

Conclusion validity. Rewording tests use 77–210 variants per kind; differences of a few points between classifiers are within noise, while the newcomer effect on the few-shot LLM is not.

7 Conclusion

Treating issue triage as a deployed, selective service changes what matters. The LLM was less accurate than a simple classifier on the benchmark but held up a year later and on a new repository; none of the classifiers could keep a promised error rate under drift without

either over-promising or deferring almost everything; harmless rewording and a newcomer's apology flipped labels, mostly among issues the service would hand off anyway; and the cheapest monitor, input novelty, was the most useful. The harness, data collection scripts, recorded model scores and regression suite are available for replication.

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